

Non-parametric regression

- Linear regression done last semester

Parametric regression. (not necessarily linear)

$$y = \alpha x_1^\beta \gamma^{x_2} \log(x_1 + \theta x_2) + \varepsilon \quad \varepsilon \sim N(0, \sigma^2)$$

the form is specified and there are finite number of unknown parameters: $(\alpha, \beta, \gamma, \theta, \sigma^2)$.

$$(x_{1i}, x_{2i}, y_i) \quad i=1, \dots, n$$

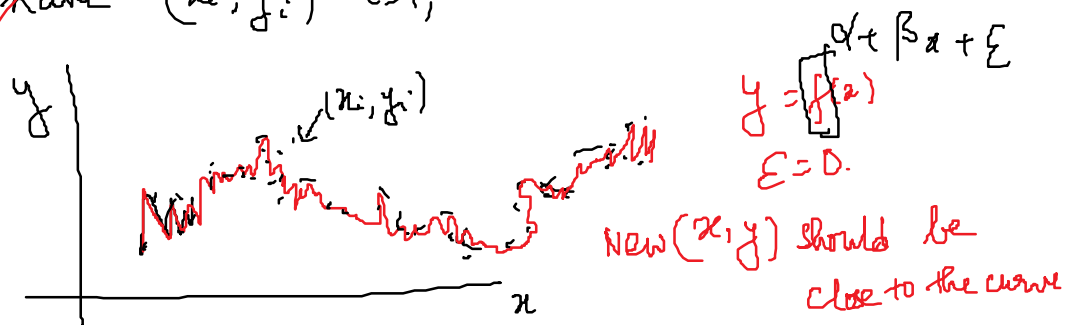
$$\min_{\alpha, \beta, \gamma, \theta} \sum_i [y_i - \alpha x_{1i}^\beta \gamma^{x_{2i}} \log(\dots)]^2 \rightarrow \text{least squares.}$$

functional

Non-parametric $\rightarrow$ Do not know the form of the dependence between x and y .

Some smoothing is done we assume that the h. $f(y = f(x) + \varepsilon)$ is "smooth". $f(x)$ is parameter of interest.

We have $(x_i, y_i) \quad i=1, \dots, n$



- Objectives
- How well the estimate $(\hat{f})$ fits the observed data
 - Complexity of the model, smoothness (number of derivatives that exist). fourth derivative is zero (eg).

We are estimating f (parameter, infinite dimensional) by

estimate $\hat{f}$ to achieve the above objectives

Least Squares estimating Y by $f(x)$

$$\text{minimize } E(Y - f(x))^2$$

$$\begin{aligned} E(Y - f(x))^2 &= E(Y - E(Y|x) + E(Y|x) - f(x))^2 \\ &= E(Y - E(Y|x))^2 + E(E(Y|x) - f(x))^2 \\ &\quad + 2E(Y - E(Y|x))(E(Y|x) - f(x)) \end{aligned}$$

$E(Y|x) - f(x)$ is non-random.

$$E(Y - E(Y|x)) = EY - E(E(Y|x)) = 0$$

The LHS is minimized as a fn of f if the second term is 0.

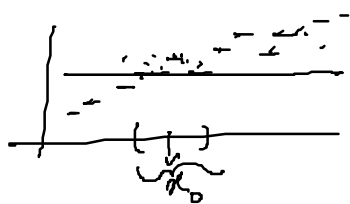
$$\boxed{f(x) = E(Y|x)}$$

$$Y = f(x) + \epsilon \quad E(\epsilon) = E(Y - f(x)) = E(Y - E(Y|x)) = 0.$$

We want the conditional expectation of Y given $X=x$.

We do not know the "distr of Y given $X=x$."

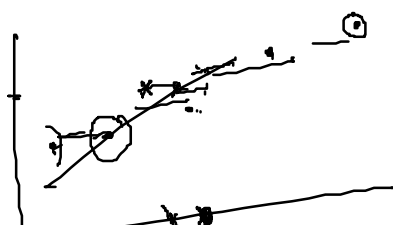
For a given x , look at $\overset{\text{many}}{k}$ x_i that are "close" to x
take the conditional distr of Y given $X=x$
as max $\frac{1}{k}$ on these corresponding y_i .

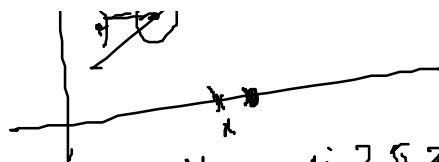


k -nearest neighbour regression.

Local averaging Piecewise st. lines that smoothes out for larger k .

$$\{y_i : |x - x_i| < h\} \rightarrow \text{Avg of those } y_i.$$





$$y = \sin 2\pi x + \varepsilon$$

↑ estimate

$$\varepsilon \sim N(0, .5^2)$$

$$\hat{y} = \hat{f}(x) = \frac{\sum w_i(x) y_i}{\sum w_i(x)}$$

$$w_i(x) = \begin{cases} 1 & \text{if } |x - x_i| < h \\ 0 & \text{otherwise} \end{cases}$$

given $X = x$, the ^{cond} distn of y

put mass $\frac{1}{k}$ to the nearest

neighbors of x .

$$w_i(x) = \begin{cases} \frac{1}{k} & \text{if } x_i \in k\text{NN of } x \\ 0 & \text{otherwise} \end{cases}$$

weights of 0 or 1 makes $\hat{f}$ rough.

Use kernel fn instead.

$$w_i(x) = \frac{1}{h} K\left(\frac{x - x_i}{h}\right) \quad (= K_h(x - x_i)).$$

$$y = y_i \quad \text{w.p.} \quad \frac{w_i(x)}{\sum_{j=1}^n w_j(x)} \rightarrow \text{cond distn}$$

$$E(y|x) = \hat{f}(x) = \frac{\sum w_i(x) y_i}{\sum w_i(x)} = \frac{\sum w_i(x) y_i}{\sum w_i(x)}$$

Bandwidth selection

$f(x) \rightarrow$ parameter

$\hat{f}(x) \rightarrow$ estimate

$$y = f(x) + \varepsilon$$

$$E(\varepsilon) = 0.$$

$$\int E(f(x) - \hat{f}(x))^2 dx \quad \text{MSE at } x.$$

y integrated.

$$\text{MSE} = \text{bias}^2 + \text{var.}$$

$$\text{bias} = O(h^2)$$

$$\text{var} = O\left(\frac{1}{nh}\right)$$

$$\text{MSE} = C_1 h^4 + C_2 \frac{1}{nh}$$

minimized at $h \propto n^{-1/5}$.

same
as
kernel/
density
estimation

minimized at $h \propto n^{-1/5}$.
 optimal rate $n^{-4/5}$

estimation

Cross validation

Fix h (Leave one out)

(x_1, y_1)

$\vdots$

(x_i, y_i) Take it out of the data

(x_n, y_n)

fit kernel regression on rest of the points

$$\sum_{i=1}^n \left(\hat{f}(x_i) - y_i \right)^2 \rightarrow \sum_{i=1}^n \hat{e}_i^2$$

$$\widehat{MSE}(h)$$

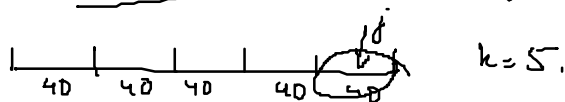
Run this over values of h on a grid.
 Find h for which $\widehat{MSE}(h)$ is minimized.

n points (x_i, y_i) , $i=1, \dots, n$. $n=200$

k fold cross validation Fix h

→ divide n data points into k equal parts.

Do this N times (100)



$k=5$.

Take out one (j^{th}) chunk of size $\lceil n/k \rceil$

$$\binom{200}{40} \binom{160}{40} \dots \binom{80}{40} \text{ many ways.}$$

$$\text{Find } \sum_{i \in \mathcal{I}^j} \left(\hat{f}(x_i) - y_i \right)^2$$

Do this for $j=1, \dots, k$ Sum.

Fix h .

Fix one division of n into k equal parts.

Take the j^{th} part. Leave it out. Estimate $\hat{f}$ based on $(k-1)$ parts.

N

$j=1, \dots, k$

$N \left[\begin{array}{l} j=1, \dots, k \\ \text{on } (k-1) \text{ parts.} \end{array} \right.$ Find the error on the j th part. by taking avg of squared errors.

$$\hat{MSE}(h)$$